

Digital Transmission of Analog Signals

Δ = Difference of Samples

- Delta Modulation (DM)

and

- Adaptive Delta Modulation (ADM)

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Delta Modulation PCM

- Delta modulation uses a single-bit PCM code to achieve digital transmission of analog signals.

- Rather than transmitting a coded representation of the sample, only a single bit is transmitted.

- This bit indicate ~~weather~~ that sample is larger or smaller than the previous sample.

Delta Modulation Process

- If the current sample is smaller than the previous sample, a logic 0 is transmitted.
- If the current sample is larger than the previous sample, a logic 1 is transmitted.

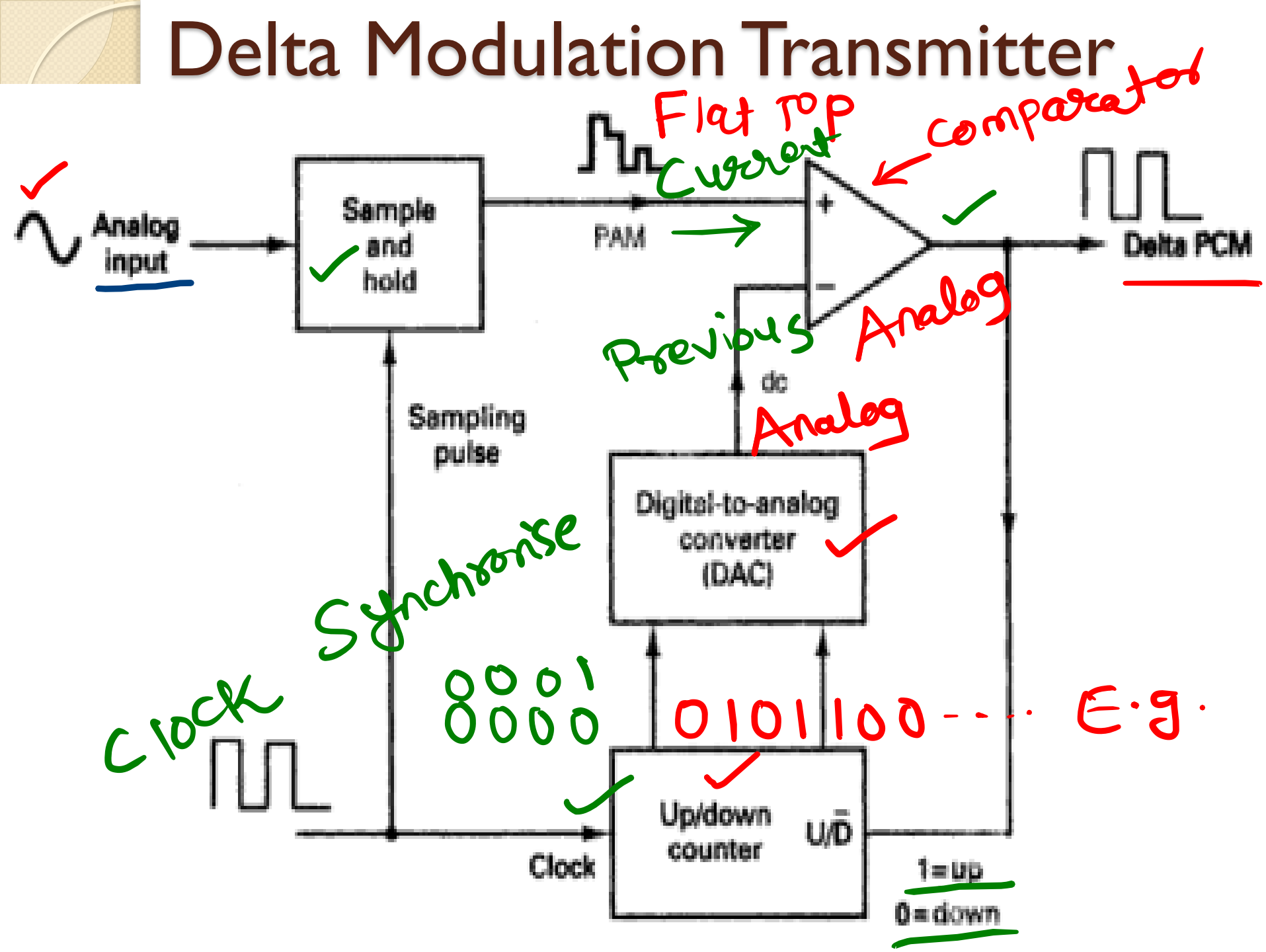
Delta Modulation Process



2 bit $\rightarrow$ 00 01 10 11 00 01 10 11

3 bit $\rightarrow$ 000 001 010 011

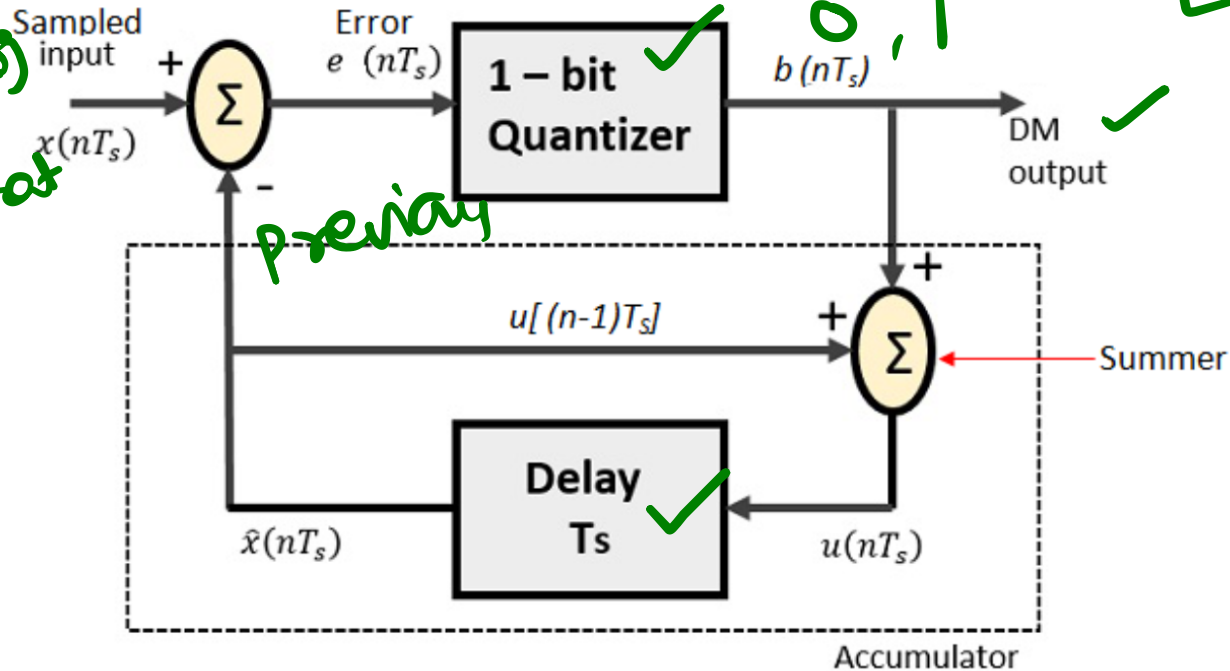
Delta Modulation Transmitter



Delta Modulation Transmitter

→ Diag. ②

Arabi
Cweat



$$u(nT_s) = u((nT_s - T_s) + [\pm\Delta])$$

or
$$u(nT_s) = u[(n-1)T_s] + b(nT_s)$$

Delta Modulation Transmitter

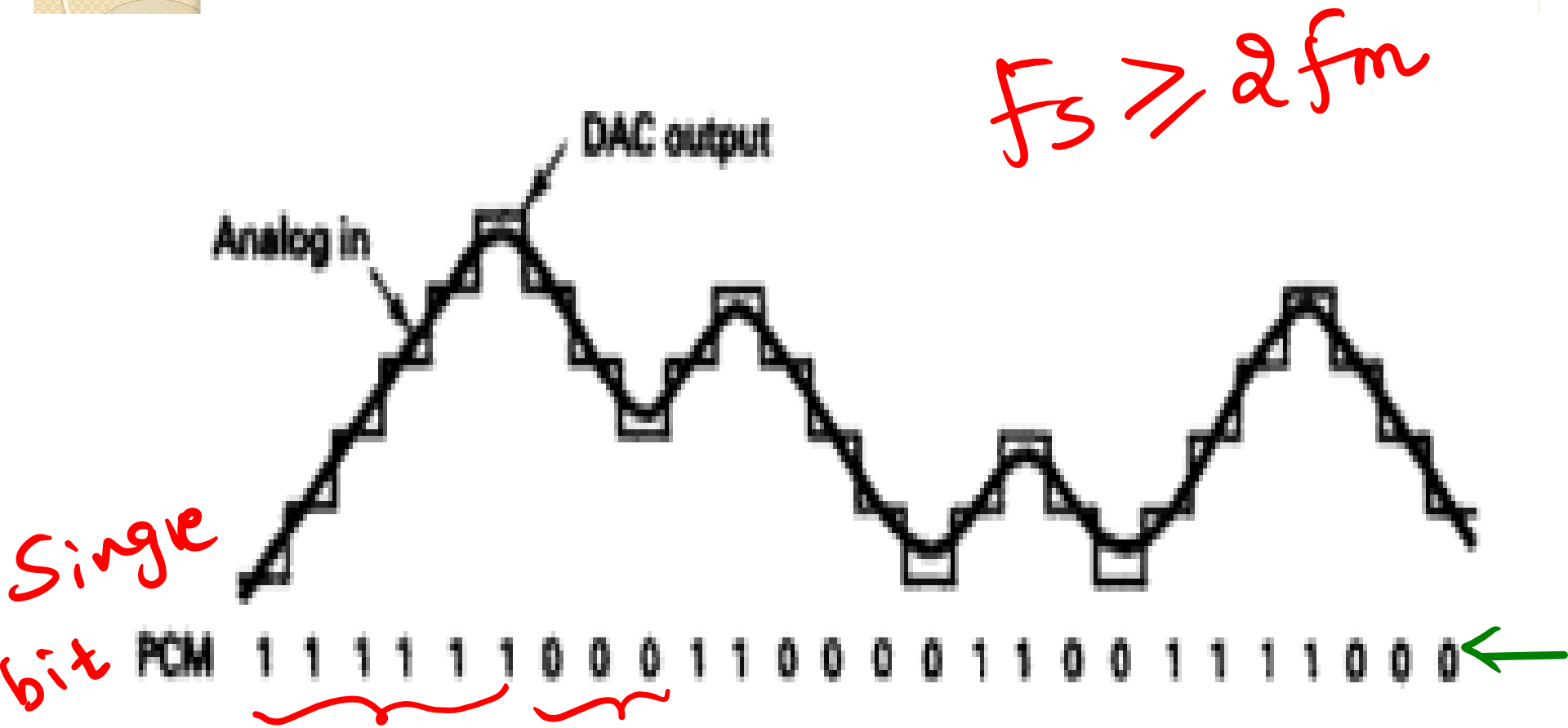
The previous sample approximation $u[(n-1)T_s]$ is restored by delaying one sample period T_s .

The samples input signal $x(nT_s)$ and staircase approximated signal $\hat{x}(nT_s)$ are subtracted to get error signal $e(nT_s)$.

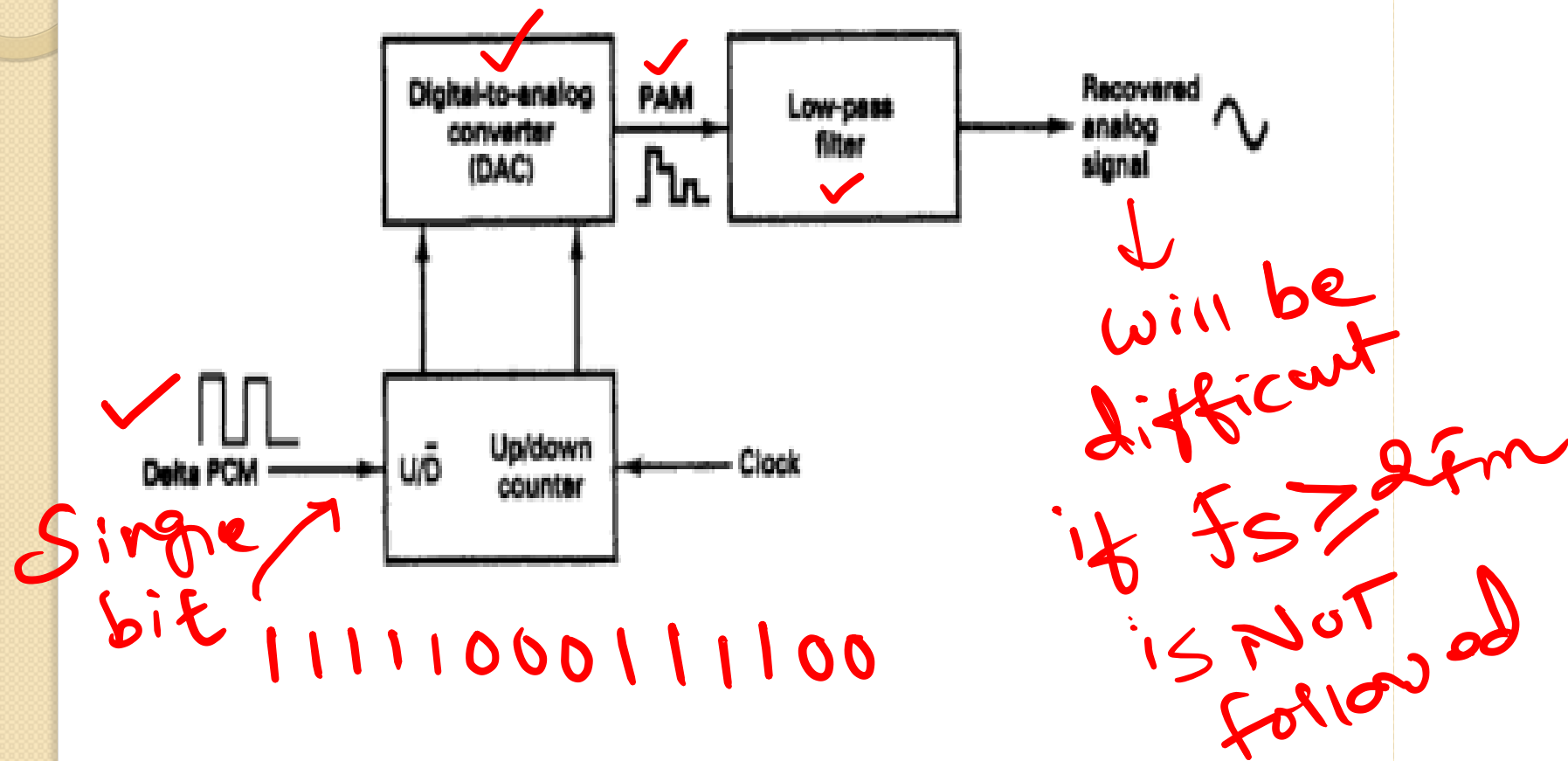
Thus, depending on the sign of $e(nT_s)$, one bit quantizer generates an output of $+\Delta$ or $-\Delta$.

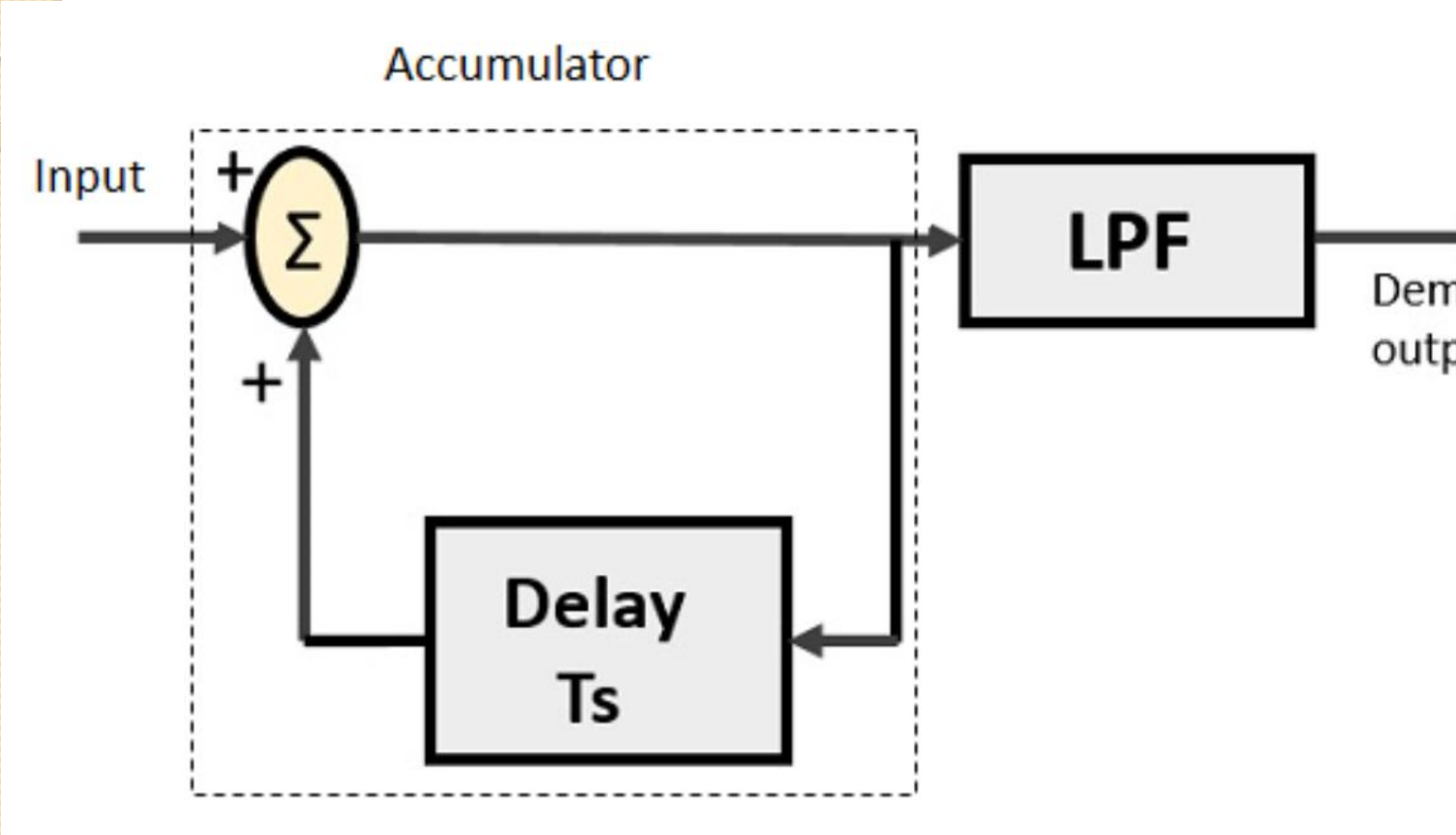
If the step size is $+\Delta$, then binary '1' is transmitted and if it is $-\Delta$, then binary '0' is transmitted.

Delta Modulation Transmitter



Delta Modulation Receiver





Delta Modulation Receiver

The accumulator generates the staircase approximated signal output and is delayed by one sampling period T_s .

It is then added to the input signal.

If the input is binary '1' then it adds $+\Delta$ step to the previous output (which is delayed).

If the input is binary '0' then one step ' Δ ' is subtracted from the delayed signal.

Also, the low pass filter smoothens the staircase signal to reconstruct the original message signal $x(t)$.

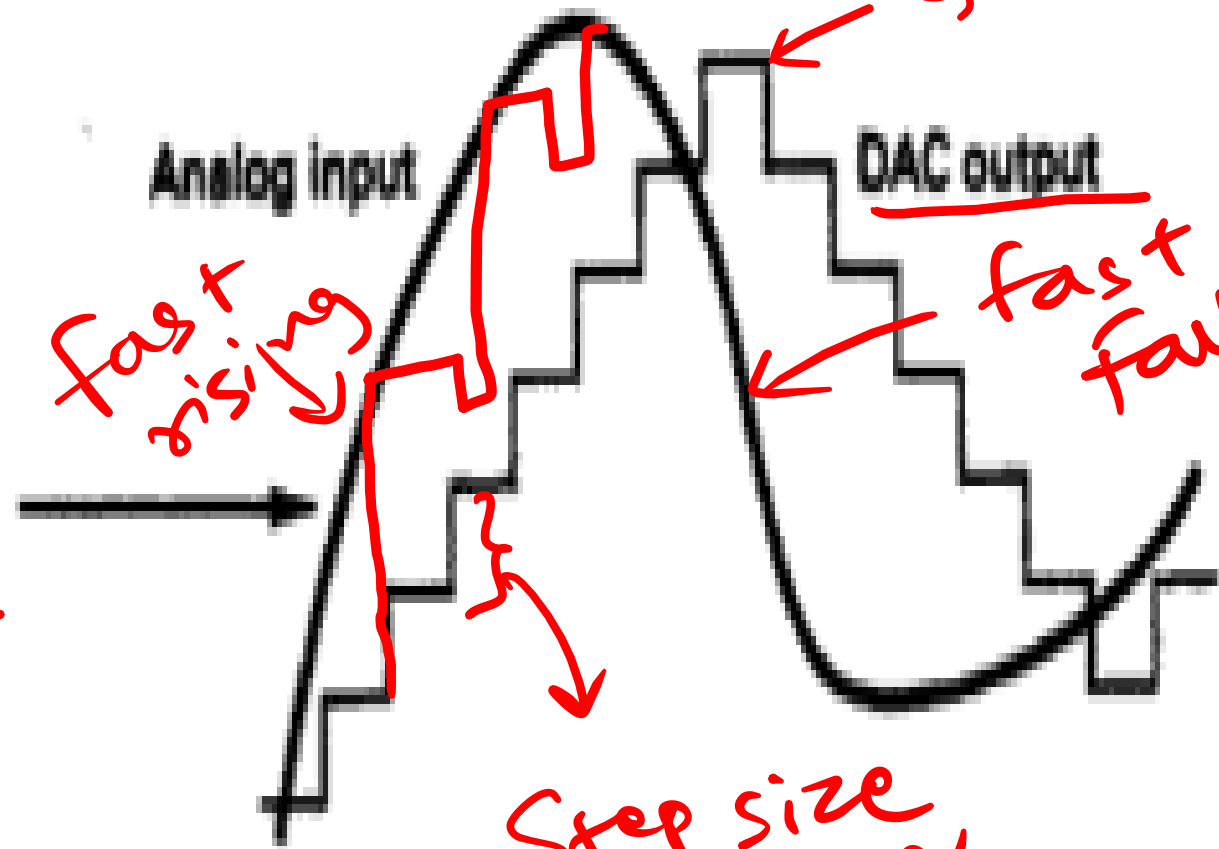
Slope overload

- If analog input signal changes at a faster rate than DAC can maintain constant output.
- The slope of the analog signal is greater than the delta modulator can maintain and is called slope overload.
- Increasing the clock frequency reduces the probability of slope overload occurring.
- ✓ Increase the magnitude of minimum step size.

Slope overload

Error
Quantized

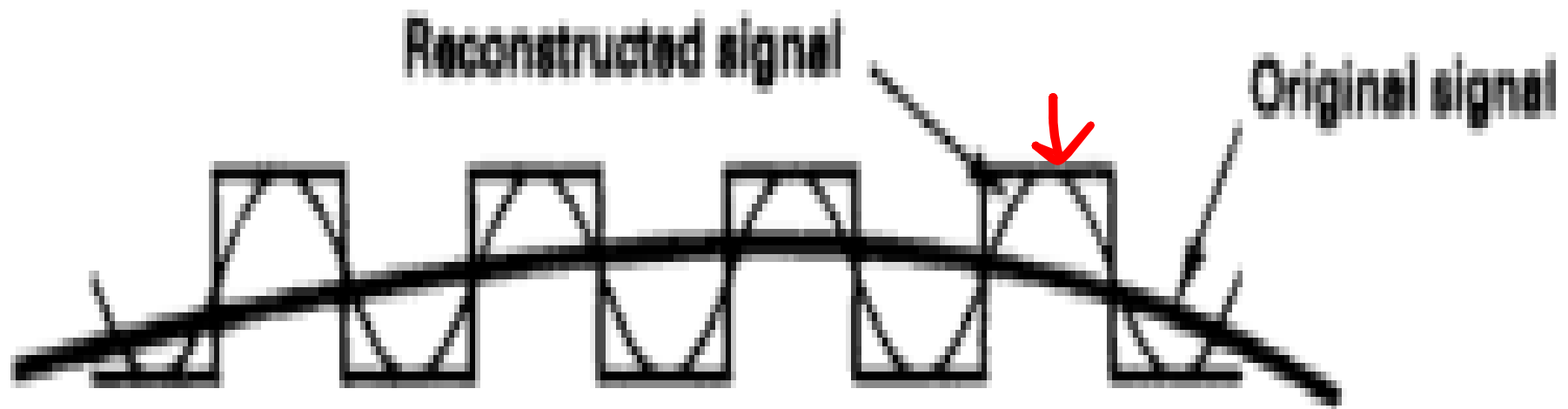
Steep slope,
rapid change



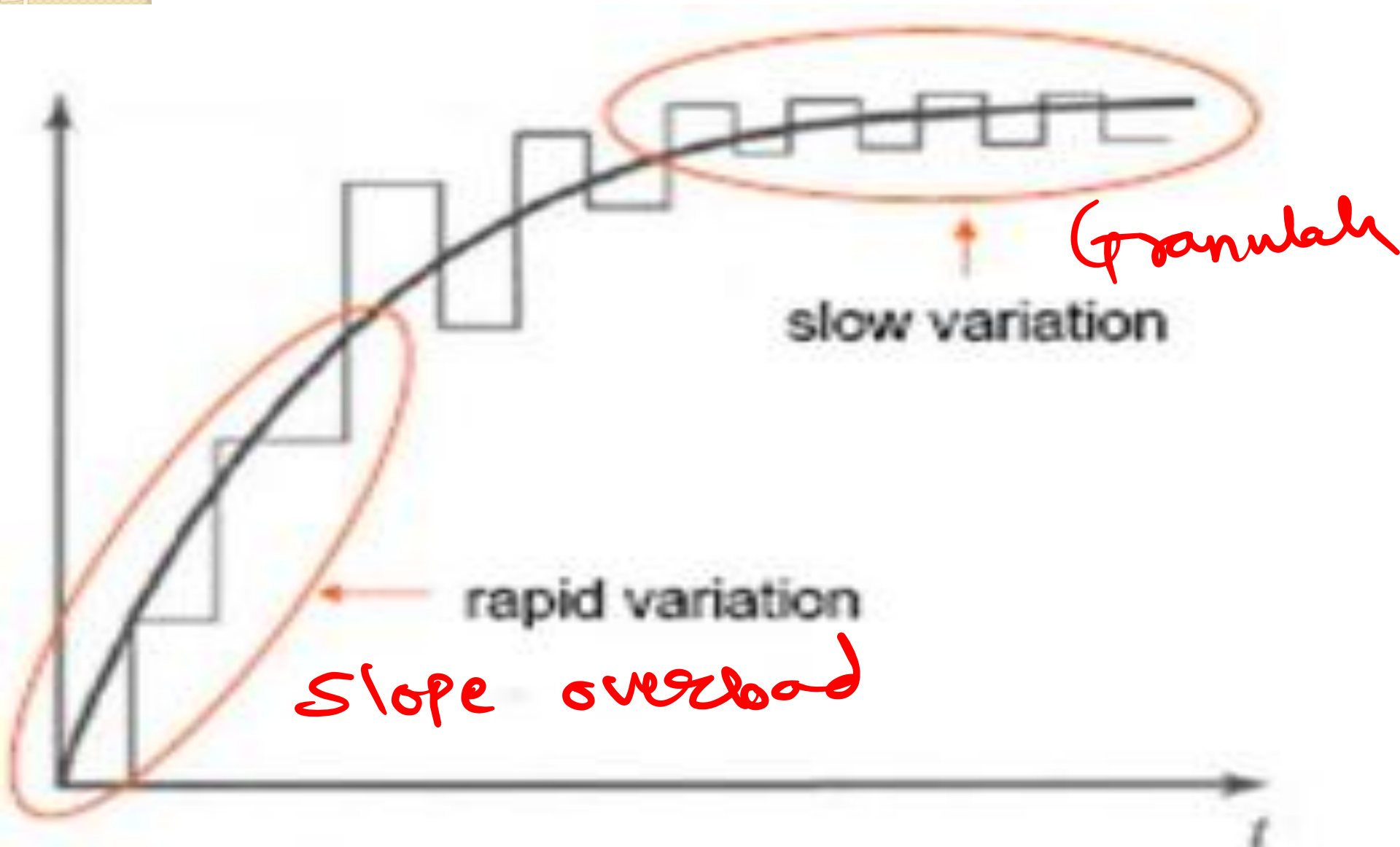
Granular Noise

- When the **original analog input signal has constant amplitude**, the reconstructed signal has variations that were not present in the original signal.
- This is called **granular noise**.
- It can be reduced by **decreasing step size**.
- Small resolution is needed to avoid granular noise.
- Large resolution is required to avoid slope overload noise.

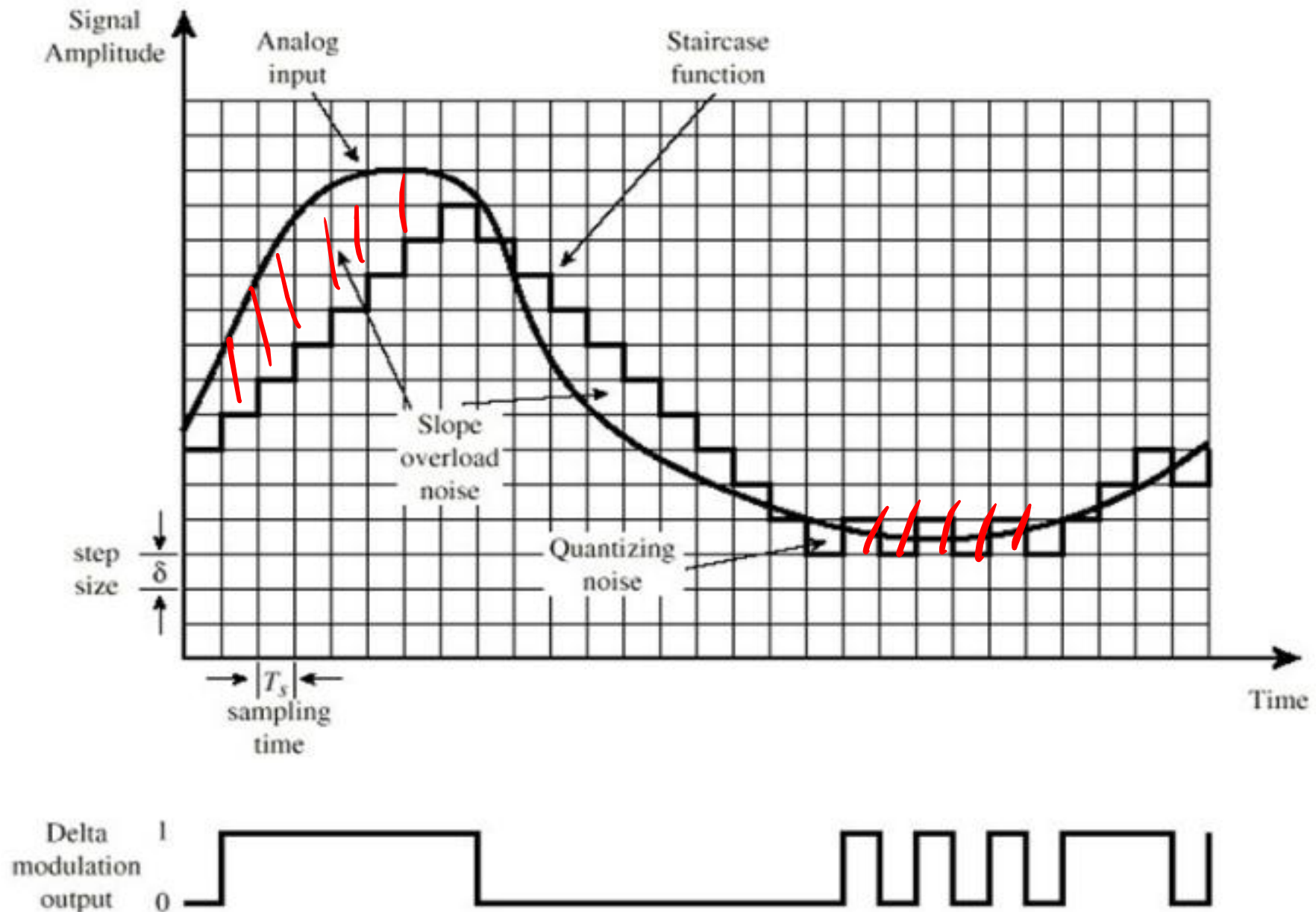
Granular Noise



Slope overload and Granular Noise



Delta Modulation Drawback



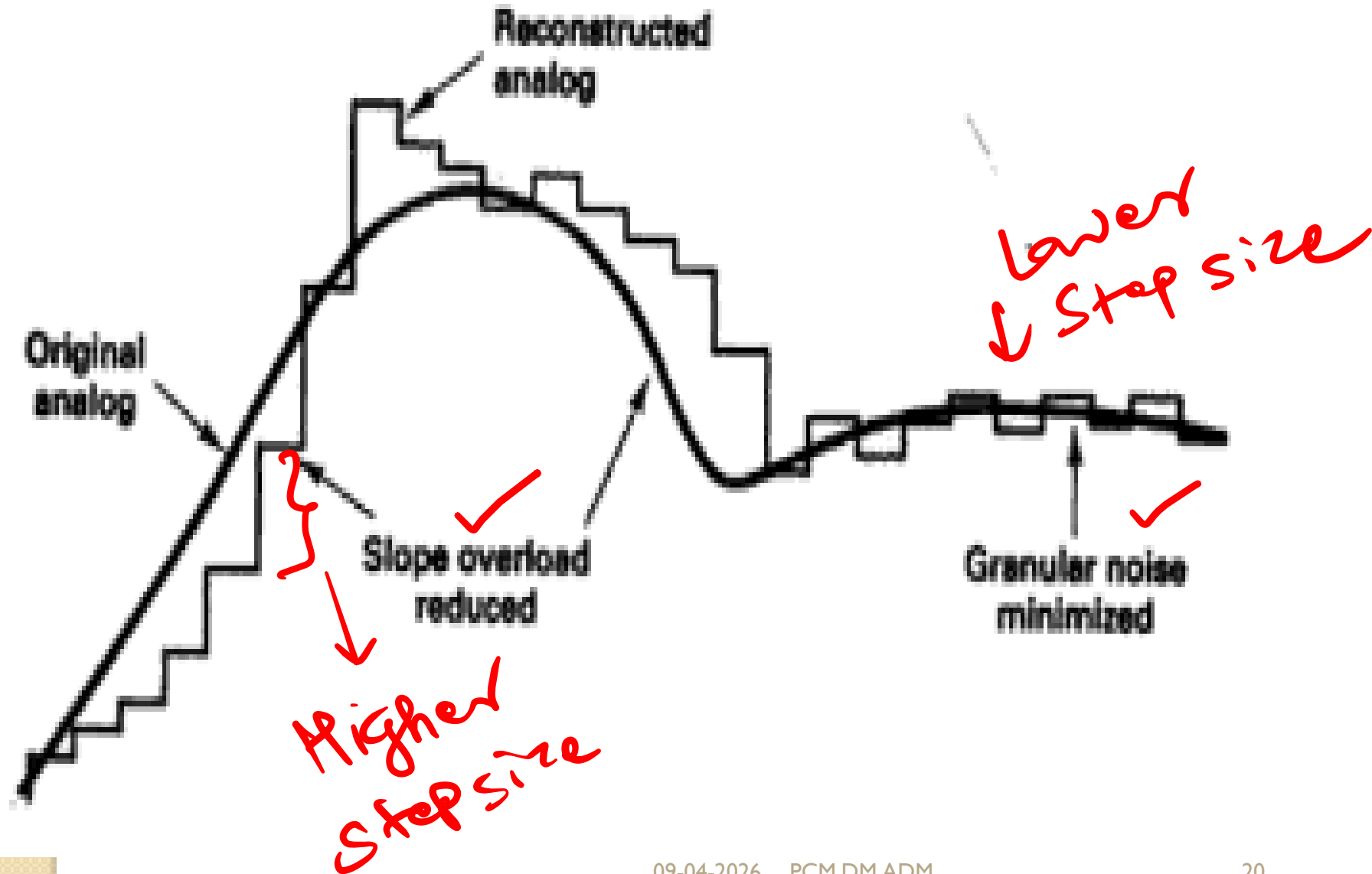
Granular Noise

- Granular noise is more in analog signals that have gradual slopes and whose amplitude vary only a small amount.
- Slope overload is more prevalent in analog signals that have steep slopes or whose amplitude vary rapidly.

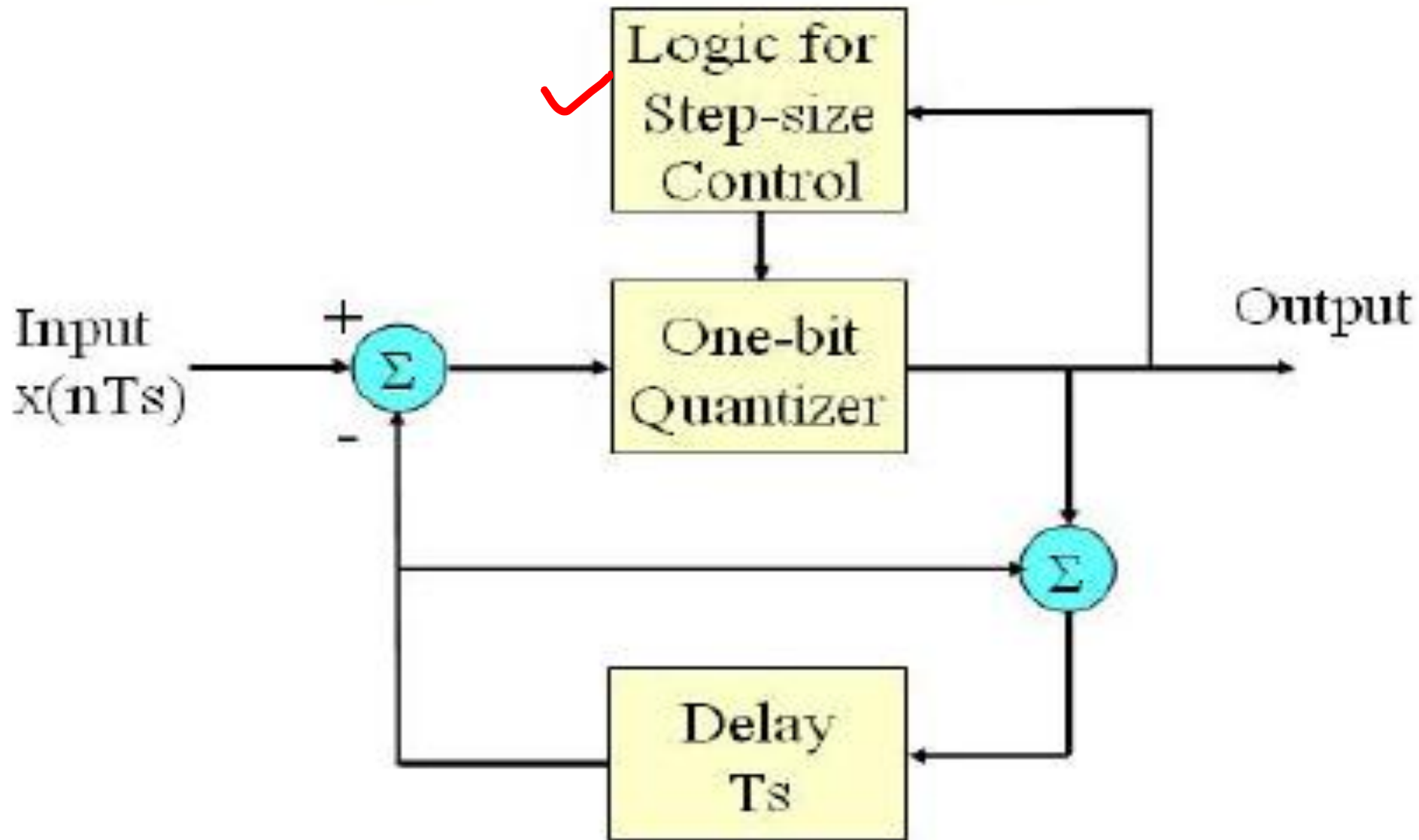
Adaptive Delta Modulation Process

- ADM is a delta modulation system where the step size of the DAC is automatically varied.
- It depends on the amplitude characteristics of analog input signal.
- After a predetermined number of consecutive 1s or 0s, the step size is automatically increased.

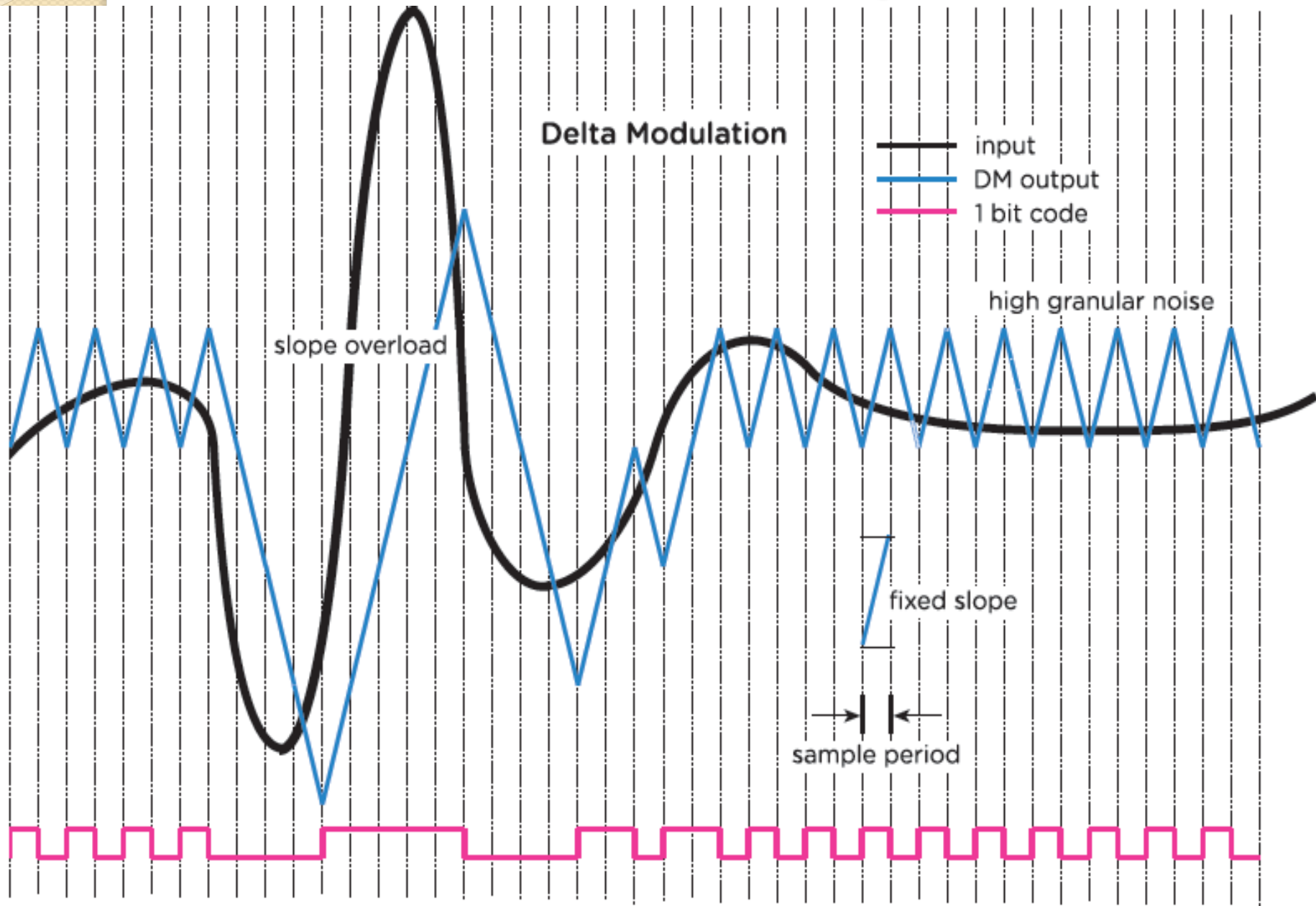
Adaptive Delta Modulation PCM



ADM Transmitter



DM Summary



ADM Summary

